

Роль 1С в цифровизации российского бизнеса

Кулясова С. А., Лошкарева Д. А., Тагайцева С. Г.

ФГБОУ ВО «Нижегородский государственный архитектурно-строительный университет»,
г. Нижний Новгород

The role of 1C in digital transformation of Russian enterprises

Digital transformation is an indispensable condition for the competitiveness of modern businesses. In Russia, given its specific legislative framework, regulatory requirements, and business practices, the «1C: Enterprise» platform has become the de facto standard for enterprise management. Its widespread adoption (over 1.5 million organizations) underscores its pivotal role in the country's economic digitalization process. Studying the role of «1C: Enterprise» allows us to understand the mechanisms and specifics of digital transformation at a national level, as well as to identify its successes and limitations [1-4].

The aim of this research is to define and analyze the role of the «1C: Enterprise» platform as the primary tool for the digital transformation of Russian enterprises, assessing its impact on improving efficiency, adapting to legislative changes, and integrating with state information systems. The main research tasks include:

- to analyze the evolution of the «1C: Enterprise» platform: from its origins as an accounting solution to a comprehensive ERP, CRM, and inventory management system;
- to evaluate the advantages of «1C: Enterprise»;
- to examine the impact of «1C: Enterprise» on key aspects of digitalization;
- to analyze the limitations and challenges of the «1C: Enterprise» platform;
- to assess the development prospects of the platform: modernization, mobile applications, machine learning, cybersecurity.

The practical significance of this paper lies in the fact that it provides a foundation for enterprises to make decisions about implementing or upgrading management systems, highlighting the advantages of «1C: Enterprise» within local realities. Understanding the platform's flexibility aids in optimizing business processes. It allows IT specialists and developers a deeper understanding of the platform's architecture and capabilities, as well as its development prospects for creating new solutions and integrations. It serves as a case study for examining the specifics of digitalization in a unique national market, showcasing the successful development of a domestic IT product [5].

Developed by the Russian software company 1C, this integrated business management system is used by over 1.5 million organizations across the country – from small retailers to large industrial firms – making it the de facto standard for enterprise software in Russia.

Originally launched in the 1990s as an accounting solution, 1C has evolved into a comprehensive platform supporting enterprise resource planning (ERP), customer relationship management (CRM), inventory and more. Its widespread adoption stems from its deep alignment with Russian legislation, tax regulations, and business practices. Unlike global ERP systems such as SAP or Oracle, which often require costly customization, 1C is natively designed for the Russian market, with built-in compliance features that automatically update to reflect changes in tax codes, reporting formats, and labor laws [6-8].

One of 1C greatest strengths is its configurability. Using a visual development environment and a high-level proprietary language, businesses and IT specialists can adapt the system to specific operational needs without extensive programming. This flexibility allows even small and medium-sized enterprises (SMEs) – which form the backbone of the Russian economy – to digitize their

workflows affordably and efficiently. Automation of routine tasks like invoicing, financial reporting, and inventory tracking reduces human error, cuts administrative costs, and frees up staff for strategic activities.

Furthermore, 1C plays a significant role in advancing Russia's broader digital economy initiatives. The platform offers native integration with major government services, including the Federal Tax Service (FTS), social funds, and the Unified State Information System, enabling businesses to submit reports and exchange data electronically with minimal manual intervention.

The platform also fosters data-driven decision-making. Modern 1C configurations include business intelligence (BI) tools that allow managers to generate real-time reports, analyze sales trends, and monitor financial performance. Integration with e-commerce platforms and banking APIs enables end-to-end digital workflows – from online orders to automated payments – improving customer experience and operational agility.

Despite its dominance, 1C faces challenges. Critics note limitations in advanced analytics, artificial intelligence capabilities, and international scalability. As Russian companies seek to expand globally or adopt Industry 4.0 technologies, they may eventually require more interoperable or sophisticated systems. However, 1C Company is actively modernizing its offerings, introducing mobile applications, machine learning integrations, and enhanced cybersecurity features [9, 10].

In conclusion, 1C is far more than an accounting tool – it is a catalyst for digital transformation across the Russian economy. By combining regulatory compliance, affordability, and adaptability with strong local support infrastructure, 1C empowers enterprises to navigate the complexities of digital modernization. In an era of rapid technological change and geopolitical uncertainty, this homegrown platform remains a vital instrument for building resilient, efficient, and future-ready Russian businesses.

References

1. Kuznetsov A.V. Tsifrovaya transformatsiya rossiyskikh predpriyatiy: rol otechestvennykh ERP-sistem [Digital transformation of Russian enterprises: the role of domestic ERP systems]. Vestnik Nizhegorodskogo universiteta im. N.I. Lobachevskogo. 2022. No 4(2). P. 145–153.
2. Lebedeva E.A., Petrov D.S. Integratsiya «1C:PredpriyatiYA» s gosudarstvennymi informatsionnymi sistemami: opyt i problemy [Integration of «1C:Enterprise» with state information systems: experience and challenges]. Ekonomika i upravlenie: nauchno-prakticheskii zhurnal. 2023. No 1. P. 67–74.
3. Mironova T.Yu. Adaptatsiya malogo i srednego biznesa k tsifrovym tekhnologiyam s ispolzovaniem platformy «1C:Predpriyatie» [Adaptation of small and medium-sized businesses to digital technologies using the «1C:Enterprise» platform]. Upravlencheskoe konsultirovanie. 2021. No 10. P. 32–36.
4. Sidorov V.I. «1C:Predpriyatie» kak instrument obespecheniya soblyudeniya zakonodatelstva pri tsifrovoy transformatsii [«1C:Enterprise» as a tool for ensuring legislative compliance in digital transformation]. Finansy i kredit. 2022. No 12(102). P. 45–51.
5. Grigoryev A.M. Arkhitektura i konfiguriruemost «1C:Predpriyatiya» kak osnovy tsifrovizatsii biznesa [Architecture and configurability of «1C:Enterprise» as a foundation for business digitalization]. Informatsionnye tekhnologii. 2023. No 3. P. 189–196.
6. Belova O.V. Tsifrovaya zrelost rossiyskikh kompaniy: rol otechestvennykh IT-resheniy [Digital maturity of Russian companies: the role of domestic IT solutions]. Vestnik RAN. 2021. Vol. 91, No 7. P. 642–648.
7. Kozlov D.A. Mobilnye prilozheniya i iskusstvennyy intellekt v «1C:Predpriyatiy»: tendentsii razvitiya [Mobile applications and AI in «1C:Enterprise»: development trends]. Problemy sovremennoi nauki i obrazovaniya. 2023. No 11(155). P. 22–27.
8. Fedorova M.S. Sravnitelnyy analiz ERP-sistem: SAP, Oracle i «1C:Predpriyatie» na rossiyskom rynke [Comparative analysis of ERP systems: SAP, Oracle and «1C:Enterprise» in the Russian market]. Ekonomika i predprinimatelstvo. 2022. No 11-2 (144). P. 110–116.
9. 1C: ITS. Digital transformation of Russian enterprises in 2023: statistics, trends, case studies. Moscow: 1C, 2023. 87 p.
10. Korotkova E.V. Kriticheskie aspekty tsifrovoy transformatsii na baze «1C:Predpriyatiya»: ograniceniya v analitike i kiberbezopasnosti [Critical aspects of digital transformation based on «1C:Enterprise»: limitations in analytics and cybersecurity]. Upravlenie biznesom. 2024. No 1. P. 55–62.